

Lab 6: AI-Based Code Completion – Classes, Loops, and Conditionals.

Task Description-1: (Classes – Data Validation)

Prompt: AI to generate a student class with attributes: name, roll-no, and marks. Add a method is pass () that returns whether the student has passed (marks $\geq$ 40).

Input:

```
class Student:
```

```
    def __init__(self, name, roll-no, marks):
```

```
        self.name = name
```

```
        self. roll-no = roll-no
```

```
        self. marks = marks
```

```
    def is pass(self):
```

```
        return self. marks >= 40
```

```
student1 = Student ("Rahul", 101, 75)
```

```
student2 = Student ("Priya", 102, 35)
```

```
print ("Student:", student1.name)
```

```
print ("Roll No:", student1.roll_no)
```

```
print ("Marks:", student1.marks)
```

```
print ("Pass:", student1.is pass ())
```

```
print (" Student:", student2.name)
```

```
print ("Roll No:", student2.roll_no)
```

```
print ("Marks:", student2.marks)
```

```
print ("Pass:", student2.ispass())
```

Output:

```
[Running] python3 -u "/Users/siddharthareddy/Documents/AI ASSISTED CODING/CRC_PROGRAM/daily.py"
Student: Rahul
Roll No: 101
Marks: 75
Pass: True
Student: Priya
Roll No: 102
Marks: 35
Pass: False

[Done] exited with code=0 in 0.061 seconds
```

Task Description-2 (Loops – Pattern Generation)

Prompt: AI to generate a function that prints a right-angled triangle star pattern using a for loop. Then regenerate the same pattern using a while loop.

Input:

```
def triangle for(n):
```

```
    for i in range (1, n + 1):
```

```
        print ("*" * i)
```

```
triangle for (5)
```

```
def triangle while (n):
```

```
    i = 1
```

```
    while i <= n:
```

```
        print ("*" * i)
```

```
        i += 1
```

```
triangle while (5)
```

Output:

```
[Running] python3 -u "/Users/siddharthareddy/Documents/AI ASSISTED CODING/CRC_PROGRAM/daily.py"
*
**
***
****
*****
*
**
***
****
*****

[Done] exited with code=0 in 0.065 seconds
```

Task Description-3 (Conditional Statements – Number Analysis)

Prompt: AI to write a function that checks whether a given number is positive, negative, or zero using if-elif-else. Test the function with multiple inputs.

Input:

```
def check_number(number):
```

```
    if number > 0:
```

```
        return "Positive"
```

```
    elif number < 0:
```

```
        return "Negative"
```

```
    else:
```

```
        return "Zero"
```

```
numbers = [10, -5, 0, 25, -12]
```

```
for number in numbers:
```

```
    result = check_number(number)
```

```
    print(number, "is", result)
```

Output:

```
[Running] python3 -u "/Users/siddharthareddy/Documents/AI ASSISTED CODING/CRC_PROGRAM/daily.py"
10 is Positive
-5 is Negative
0 is Zero
25 is Positive
-12 is Negative

[Done] exited with code=0 in 0.06 seconds
```

Task Description-4 (Nested Conditionals)

- **Prompt:** Generate a function `check_discount(age, is_member)` that determines discount eligibility:
- **Age ≥ 60 $\rightarrow$ Senior discount**
- **Member $\rightarrow$ Additional discount**
Use nested if statements.

Input:

```
def check_discount(age, is_member):
```

```
if age >= 60:
    print("Senior discount eligible")

    if is_member:
        print("Additional member discount")
    else:
        print("No additional member discount")

else:
    print("Not eligible for senior discount")

    if is_member:
        print("Member discount eligible")
    else:
        print("No discount")
```

```
print("Customer 1:")
check_discount(65, True)
```

```
print("\nCustomer 2:")
check_discount(70, False)
```

```
print("\nCustomer 3:")
check_discount(35, True)
```

```
print("\nCustomer 4:")
check_discount(30, False)
```

Output:

```
[Running] python3 -u "/Users/siddharthareddy/Documents/AI ASSISTED CODING/CRC_PROGRAM/daily.py"
Customer 1:
Senior discount eligible
Additional member discount

Customer 2:
Senior discount eligible
No additional member discount

Customer 3:
Not eligible for senior discount
Member discount eligible

Customer 4:
Not eligible for senior discount
No discount

[Done] exited with code=0 in 0.091 seconds
```

Task Description-5 (Class – Mathematical Opera)

Prompt: Ask AI to create a Circle class with methods to calculate area () and circumference () given the radius.

Input:

```
import math
```

```
class Circle:
```

```
    def __init__(self, radius):
```

```
        if radius < 0:
```

```
            raise ValueError("Radius cannot be negative")
```

```
        self.radius = radius
```

```
    def area(self):
```

```
        return math.pi * self.radius ** 2
```

```
    def circumference(self):
```

```
        return 2 * math.pi * self.radius
```

```
    def __str__(self):
```

```
        return f"Circle(radius={self.radius})"
```

```
c = Circle(5)
```

```
print(f"Radius: {c.radius}")  
print(f"Area: {c.area():.2f}")  
print(f"Circumference: {c.circumference():.2f}")
```

Output:

```
[Running] python3 -u "/Users/siddharthareddy/Documents/AI_ASSISTED_CODING/CRC_PROGRAM/daily.py"  
Radius: 5  
Area: 78.54  
Circumference: 31.42  
  
[Done] exited with code=0 in 0.071 seconds
```